

Universal Cake -- An Introduction

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Abstract

This document introduces Universal Cake, a framework for thinking about technology in human terms. It explains where the idea came from, what the name means, and how it can help anyone, technical or not, ask better questions about the tools they use every day.

Inspirations

Universal Cake began in 1999 as a personal checklist. Christopher Steel was looking for web software and wrote down the questions that he thought mattered the most to his community: Does it come in my language? Does it work on my system? Does it work in the browser I use? Is it accessible to people with disabilities? Is it supported? Is it maintained? Does it have the qualities needed for longevity? Is it worth investing time and energy in?

That list was not borrowed from anywhere. It came from experience, from working with real people using the technology available at the time.

Over time, the questions on that checklist changed with many changes in technology, the advent of better approaches to making browsers compatible and the rest of the evolving technology landscape. The list was broken down into sections, some items overlapped multiple sections, it continued to be adapt and change and today has become more of a framework, or perhaps even a praxis.

During this time Christopher began to encounter others who had been asking similar probing questions from their own technological directions and evaluating not only tech, but other subject. Some he was not even aware of until recently, but it has become clear that a lot of parallel thinking emerged in many different disciplines since that initial list was developed, and some prior work that he was not aware of also share similar approaches to the thinking that went into his framework which was growing into a very useful tool in selecting what tech Christopher would invest his resources in. The framework was now a very valuable tool for Christopher and was becoming something pretty special. Almost a personal oracle of sorts.

For example, [Meadows \(2008\)](#) and her book *Thinking in Systems* was a recent and satisfying discovery: the idea that healthy systems are made of interdependent parts that support one another maps well with the framework that Christopher was using for his evolving recipes designed to create a better Universal Cake.

Meadows, original manuscript, written in 1993, was circulated informally for years before publication in 2008, meaning Meadows had already been working through similar ideas regarding systems thinking for years by the time Christopher made his first one-page checklist. A researcher had mentioned that Christopher might be interested in her work sometime in the late 2010s. When he encountered her work again doing research in 2026, he took a deeper dive and discovered that the researcher was right, her ideas had a lot in common with his approach to creating Universal Cake and perhaps her ideas could help transform Universal Cake into something even more effective.

The human inspiration behind Universal Cake is wide and does not fit neatly into a reference list. It spans disability rights work, including the research of [Jutta Treviranus at OCAD University](#), the ground-breaking real-world integration of inclusion by [Catherine Roy, Accessibility Advisor for the McGill University Equity Team](#) (<https://www.mcgill.ca/equity/catherine-roy>), and the advocacy of organisations such as [RAPLIQ](#) and [Vie Autonome Montréal](#) (<https://va-m.org/>). These ideas and initiatives, alongside philosophy, systems thinking, trauma-informed practices, and personal resilience help transform Universal Cake. People like Alex Limi and the Plone community, Hugh Gallagher, Massimo Di Pierro, bell hooks, Donella Meadows, Pema Chödrön, Ed Roberts, Michael DeHaan, and many others have all shaped how Christopher thought and still thinks about people, systems, and what it means to build something that serves rather than excludes. Technology does not exist apart from these questions. It is shaped by them, and shapes them in return and is an important relationship in determining our individual and collective wellbeing.

The plain language principles used in this document are guided, in part by the [Plain Language Action and Information Network \(2011\)](#).

1. Where it all started

In 1999, Christopher Steel wanted to select a website software that allowed for online editing from thousands of potential candidate using the [AltaVista](https://en.wikipedia.org/wiki/AltaVista) (<https://en.wikipedia.org/wiki/AltaVista>) web search engine. At the time thousands of projects existed. In fact this was the year that the highly successful Blogger was launched on **August 23, 1999** by Evan Williams and Meg Hourihan through their company Pyra Labs. Many other coders were attempting to create similar products. Chris was looking for the right project that would work for a community that included people with disabilities, people who did not read or write English, some of whom were Francophone and some who were using older hardware and operating systems. He began compiling a list of qualities that he thought mattered the most for this community with the idea that success would depend on how useful to tool would be for end users. As his list of desirable qualities grew into something much larger he decided that he needed a name for this growing collection of needs, something that would help him think more clearly about his criteria in ways that would support his community and his personal priorities.

The name came from an unlikely place. A few years earlier, Christopher and some artist friends had started a communications company. They came up with the name Universal Cake, designed an actual box for the product, and then could never quite figure out what should go in it. The company did not survive. The name did. It fit his growing list perfectly, and the rest is history.

Accessibility moved to the top of that list not as a political statement but as a logical one. If the software did not work in your language, or on your system, or for your users, then nothing else on the list mattered. Speed, stability, and sustainability were all important, but they were downstream of access. Without the door, the rest of the house is irrelevant.

Evaluating the options against the list, Christopher found Plone. It was not finished. But its commitments were clear: multilingual, open source, web-editable, with accessibility treated as a first-class concern rather than an afterthought. He chose it, and he waited for it to mature. That turned out to be a good investment. Plone is still here. Most of the software that competed with it in 1999 and 2000 is not. The list worked.

Plone's ambitions at the time seemed outrageous to many. No one had done this at scale in open source before, a web-based editor that anyone could use without technical training, in over fifty languages, built from the ground up with accessibility as a first-class concern rather than a feature added later. Out-of-the-box Plone installations in the early 2000s were documented as meeting WCAG 1.0 AAA, the highest level under the standard in force at the time, making Plone one of the very few mainstream content management systems to claim that distinction.

WCAG 1.0 AAA meant satisfying all Priority 1, 2, and 3 checkpoints, which was achievable for a well-built default theme in that era. When WCAG 2.0 arrived in 2008 with a substantially revised structure and more rigorous requirements, the bar changed, but Plone's commitment did not.

Plone's design went on to influence Wikipedia's MonoBook skin, which was the default face of the world's largest encyclopedia for six years. The organisations that eventually adopted Plone, the United Nations, the FBI, and governments on multiple continents, had exactly the same requirements as that original checklist. They needed systems that worked across languages, across abilities, and across decades. The list was not unusual. It was just an honest assessment of the real-world human needs and requirements that all humans have for technology that serves them.

2. What the name means

The word *universal* describes the goal: technology that works for the widest possible range of people, languages, abilities, devices, and circumstances. Not technology designed for what is wrongly described as the comfortable 80% at the centre of the bell curve, but technology that keeps asking, in the words of [Treviranus \(2019\)](#), "who are we missing?", Universal Cake works hard to find real answers to this question.

[Treviranus \(2019\)](#) argues that the conventional reading of Pareto tells developers and designers to serve the easy 80% and ignore the "difficult 20%" at the margins. This is both a misreading of Pareto and that misunderstanding has led to a wrong turn in many of our approaches, in particular. It has become a "standard practice" in many tech shops and has huge impacts on the suitability of much technology that we use today in failing to support human agency and wellbeing.

Where true innovation and knowledge lie

Treviranus points out that the people at the margins, the vital few, actually cover 80% of the knowledge terrain. Designing for them does not cost you. It repays you in knowledge and understanding, and in systems that reach the edges of human diversity.

This is the very space in which innovation, resiliency, and significantly better systems that serve the real population are born. The dynamic that results from this commitment, a continuous, iterative process of asking who we are missing and expanding inclusive design practices to reach them, creates systems, like Plone, that grow stronger and more capable with each cycle. Treviranus describes this effect as the Virtuous Tornado.

Christopher, and many others, were working with similar intuitions in 1999 and before, but Treviranus put a highly descriptive name on the results of these methods. The convergence is not a coincidence. It is what happens when people think honestly about similar problems within the context of our real world and the real people, creatures and plants that inhabit it.

2.1 The 80/20 rule is not a law, and it was never about you

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It is worth pausing on the 80/20 rule, because it has done enormous damage in the name of common sense, most especially in the field of software development.

Vilfredo Pareto was an Italian economist working in the 1890s. His original observation was very specific: approximately 80% of Italy's land was owned by 20% of its population ([as cited in Wikipedia, 2024](#)). Ironically, as [Treviranus \(2019\)](#) notes, Pareto had nothing to do with the development of the term that would come to bear his name. Literally nothing at all other than the percentages 80% and 20%.

Pareto's observation was about wealth concentration in one country at one moment in history. Even that basic statistic does not hold true in the world today. It is not true, it is not some magic formula, it is now a universal law of nature. It was not even a design principle and it was not an instruction or a rule.

Yet it, inadvertently, it became all three to many of the individuals responsible for creating the code that runs many of the services we rely on. Applied to technological design, the 80/20 rule became a licence to ignore anyone who was not already well-served, to treat a comfortable majority as the only audience worth building for, and to call that efficiency and effectiveness.

The problem is that the 80% principle (as implemented in the field of technology) describes was never the world's 80%. It was the 80% of a very specific and narrow market, usually composed of English-speaking consumers, with high bandwidth and in possession of recent hardware, that was considered to be the included ones before a technological project even starts.

The actual numbers tell a different story. Around 380 million people speak English as their native language, roughly 5% of the world's population ([Eberhard et al., 2024](#)). Even including everyone who speaks English as a second language, more than 4 in 5 people in the world do not understand English, yet 54% of websites worldwide are in English ([Statista, 2024](#)). The technology industry did not design for 80% of the world. It designed for a very small minority of it, gave that minority a name, "the majority", and then applied a rule about Italian land ownership in the 1800's in order to justify ignoring everyone else.

Today there are some 7,170 living languages in the world. Roughly 44% are now endangered, often with fewer than 1,000 users remaining ([Eberhard et al., 2024](#)). Every one of those languages represents a community of people, a way of understanding the world, a body of knowledge that exists nowhere else. The 80/20 rule, as applied to technology, has nothing to say about any of them. It cannot even see them.

The idea of having a target market is not wrong in every context. Applied carefully, within a clearly defined and honestly described market, it can be used as a tool for prioritisation. But it requires honesty about who that market actually is, and who it is not. When the target market is "English speakers on modern hardware with fast internet connections," the rule describes that market. It says nothing about the majority of humanity other than we do not acknowledge or see you.

Universal Cake does not pretend the 80/20 rule does not exist. It insists on asking: 80% of whom, exactly? And what happens if we leave out everyone else? Is this efficiency? Is this effectiveness, helpful? or even sustainable? Is this how we are going to invest our very limited stocks?

2.2 When every product ignores what they consider to be the same 20%, the same people lose everything

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There is a second major blind spot of the 80/20 "rule" which is frequently interpreted as the difficult 20%, It is more difficult to grasp but significantly more damaging in practice.

It is not just that any one product ignores 20% of its potential users. It is what happens when every product does this independently, each to its own version of the "majority."

Consider a person who speaks a minority language, lives in a low-income household, uses older hardware, has low or intermittent bandwidth, and has a visual impairment. Each of these characteristics places them outside the target market of a different set of products and services. The ATM, the banking app designed for smartphone users on current hardware. The government portal that is only available only in the dominant language. The healthcare platform that requires a screen resolution that older device cannot produce. The job search tool assumes broadband speeds which are not available in the neighbourhood.

Each product, in isolation, made a reasonable-sounding decision to serve its majority. But the cumulative effect on this one person is not a 20% reduction in access. It is near-total exclusion from the digital infrastructure upon which modern life increasingly depends.

Researchers call this **compounded digital exclusion**, the phenomenon in which digital inequalities do not operate along independent axes of division, but overlap, interlink and interact, producing outcomes far more severe than any single exclusion would suggest ([Zheng & Walsham, 2021](#)). When the 80/20 rule is applied independently across thousands of products and services, each drawing its target market from roughly the same centre of the bell curve, the people at the edges are not excluded from one thing. They are excluded from most things. The exclusions stack and these populations are frequently the targets of cost cutting initiatives designed to avoid disrupting the privileged.

This is not a marginal problem. The people most likely to be at the edges of multiple target markets simultaneously are also the people who most depend on technology for access to vital services, healthcare, education, employment, government, banking, communication. The 80/20 rule, applied at scale across an entire technology ecosystem, does not produce a world where 80% of people are well-served and 20% are somewhat underserved. It produces a world where the same people are comprehensively excluded from nearly everything, while the comfortable centre is served by an abundance of competing options.

2.3 The cost of ignoring the edge

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There story from community organising that illustrates the cost of ignoring the edge. Christopher once canvassed door to door for an environmental organisation. When the question came up of how to handle donors who already gave to other environmental groups, considered competitors, in the conventional sense, the answer from organiser Kevin Kintner was not what most organisations or door to door canvassers might have expected.

Other groups talked about dividing the pie fairly, how to share a fixed pool of donors between organisations working in the same movement. Kintner's answer was very different. The goal, he argued, was not to divide the pie. The goal was to create a bigger pie. When a canvasser encountered someone who already contributed to another environmental organisation, the right response was actually to thank them, genuinely, for supporting the movement, and then to talk honestly about what our organisation did, how it worked, and what made it distinct. No competition. No zero-sum thinking. Just more information, more choice, and more trust communicated to the many supporter of the movement.

The result was that donors could discover which organisations most closely matched their own values and commitments. Some gave to multiple groups. Some switched groups. Some increased their giving overall. The movement continued to expand and grow because the approach expanded the number of people who saw themselves as participants in a movement rather than salespeople seeking easy sales.

This is exactly what [March \(1991\)](#) describes as the difference between exploitation and exploration, and what [Putnam \(2000\)](#) identifies as the distinction between bonding capital, strengthening existing relationships, and bridging capital, building new relationships across social boundaries. The organisations that only work the "known turf" eventually run out of it. The ones that invest in the edges, in people not yet engaged, are the ones that grow and benefit from the resilience that is found only at the edge.

The same pattern appears in ecology. An **ecotone** is the transitional zone between two ecosystems, the place where forest meets meadow, where river meets land. [Odum \(1953\)](#) first described what he called the edge effect: increased species richness and abundance in boundary regions, along with the presence of species found nowhere else. Species living in ecotones are constantly adapting to the dynamic conditions of life in the in-between, which makes them especially resilient to change. The comfortable centre of any ecosystem is well-served and well-optimised. It is also the most fragile. The edges have been asked to cope. They have developed capabilities the centre never needed.

The same holds true for organizations, many of which concentrate power in the form of individuals sometimes defined as the elite. The individuals who are the least knowledgeable about what it means to live on the edge and the individuals who are the least practices in dealing the traumatic change.

One of the more unexpected confirmations of this idea in human intellectual life comes from the economist Thorstein Veblen. In 1919, [Veblen \(1919\)](#) attempted to understand why Jewish thinkers had made such disproportionate contributions to European intellectual life. His argument was not about inherent characteristics. It was about position, what he called hyphenate status: living simultaneously between two communities, belonging fully to neither. That position, he argued, produced a particular quality of critical and creative thought, the capacity to see from outside what those fully inside cannot question. His research did not fully succeed on its own terms and the essay has been largely set aside. But the core observation has not aged badly. It is the ecotone argument applied to human minds.

The current research on human edge communities, runs in two distinct directions simultaneously, and the honest version of this argument must hold both.

[Meyer \(2003\)](#) formalized what is now known as the minority stress model: the chronic, socially-based stress experienced by marginalized groups through stigma, discrimination, and exclusion produces measurable, serious, and well-documented health disparities. Historically marginalized populations incur a higher risk of developing chronic diseases across the course of life, with emerging evidence showing that the biopsychosocial effects of social stress shape rates of biological aging

through the inflammatory process ([Brown et al., 2020](#)). Living at the edge, under conditions of exclusion, makes people sick. That is not metaphor. It is measurable physiology.

The economic costs are equally well documented. [Buckup \(2009\)](#) for the International Labour Organization calculated that economic losses related to the exclusion of persons with disabilities from the labour force are large and measurable, ranging from between 3 and 7 per cent of GDP. Critically, people with disabilities are less productive not because they are "disabled" but because they live and work in environments that are "disabling."

The cost is not in the people. **It is in the systems that exclude them.**

Beyond the direct economic losses, the social and political costs compound: exclusion narrows the range of voices shaping decisions, concentrates power in the centre, and produces systems that serve an ever-smaller slice of the population while accumulating the costs of everyone left out.

And yet, resilience is not merely the absence of dysfunction following adversity, but rather an engagement with life that promotes personal growth and a deeper understanding of oneself and others. [Tedeschi and Calhoun \(1996\)](#) identified what they called post-traumatic growth: the positive psychological changes that can emerge from the struggle with significant adversity, new understandings of self and world, stronger relationships, a heightened appreciation for life, and capabilities that would never have developed without this challenge.

Research on multicultural experience confirms a related pattern: people who have had to navigate between worlds develop greater idea receptiveness and creative potential than those who have inhabited only one ([Leung & Chiu, 2010](#)).

Edge communities develop these capacities out of necessity. They learn to adapt, to read systems from the outside, to find solutions in the gaps between what was designed for them and what they actually need.

This is not a reason to celebrate exclusion or to treat suffering as a productive condition.

There is a real danger in pointing to the resilience and creativity of edge communities without acknowledging what produced them, and without taking responsibility for changing the conditions that make those qualities necessary. The growth is real. The harm is real and the harm that preceded the growth was not inevitable or acceptable.

In the study of epizootic's (mass die offs of animals, corresponding to epidemic diseases in humans) the science of fish stress offers one of the clearest illustrations of why systemic conditions matter, and why they are so difficult to address through conventional legal and regulatory frameworks. [Sindermann \(1996\)](#) documented how environmental stressors, pollution, overcrowding, temperature change, habitat degradation, do not kill fish directly. They compromise immune function. They weaken the animal's capacity to resist what it would otherwise survive. The fish then dies, officially of a pathogen, a parasite, or an infection that a healthy fish would have fought off without difficulty. The interpretation could be made that a particular toxin or general pollution or did not kill the fish, but it up the conditions in which something else could kill the fish.

This creates a profound problem for environmental law. In the US Environmental claimants face an enormous burden in establishing both that a defendant's pollution is capable of causing the alleged injury, and that it was a substantial contributor in this particular case ([Harvard Law Review, 2015](#)). You can sue the company that dumped the toxins. You cannot easily prove that those toxins, rather than the pathogen that delivered the final blow, caused the population to fail. The legal system requires a proximate cause. Living systems do not work that way. Cumulative stress creates conditions. Something opportunistic exploits them. The thing that set up the conditions walks free.

The same logic runs in the opposite direction, and this is where the analogy becomes something more than ecological. If you cannot point to the specific pollutant that killed the fish, you equally cannot point to the specific intervention that would save it.

Wellbeing, in fish, in humans, in communities, is not produced by a single input. B12 supports stress resilience, but it does not clean the water. A change in socioeconomic status may eventually reduce chronic stress, but the transition itself introduces new pressures, and the benefits, are slow and diffuse to take hold. Even a dramatic improvement in circumstances does not immediately produce wellbeing, because the body and mind have been calibrated to survive in different conditions, acclimatisation to a new environment, even an improved one and the recalibration following acclimatisation takes time and in some cases, a lot of work.

What improves wellbeing is the same thing that improves the fish: cleaner water, sustained across time, across the whole system. Not a single intervention. Not one policy. Not one feature added to one product. The cumulative quality of all the conditions in which a life is lived, our relationships with our body, our minds, and our world that either support or degrade our capacity to flourish are what determine an embodied sense of wellbeing. This is slow, systemic, and sometimes unglamorous work.

What matters is the quality of the work as a whole. A beautiful door does not produce a beautiful home, but it helps. Being well does not produce a single attributable cause of improvement any more than pollution produces a single attributable cause of harm. Wellbeing might be better measured as the sum of the quality of our relationships with ourselves and our world but that does not explain the positive power or the virtuous tornado, nor the extreme consequences of compounded exclusion of any type.

Working on raising the qualities of our relationships is the only kind of work that actually changes the conditions rather than managing the symptoms

In other words, if you are lucky enough to be able to pay for a prescribed some of the most powerful medicines, for example, **Disease-modifying antirheumatic drugs (DMARDs)**, these might give you relatively quick relief, but like most drugs you would need to take it for the rest of your life in order to benefit from that relief. Working on the quality of your relationship with yourself and others is also something that can be done on a daily basis, but in this case, the "drug" is free and the results are cumulative and holistic in nature.

This is what Universal Cake is asking for. Not a feature. Not a fix, but a sustained commitment to clean water.

2.4 Therefore...

Universal Cake asks designers, developers, and decision-makers to hold this full picture in mind, not just the target market for a particular product, but the full picture, including the cumulative effect that target markets have on the people who fall outside most of them.

The word *cake* describes something everyone loves. A cake is made of ingredients. No single ingredient makes a cake, it takes all of them, in the right relationship with one another in order to produce something delicious and well worth eating.

Universal Cake uses this image to describe how good technology is created: not as a single feature or a single metric, but as many qualities working together as a whole. Those qualities include accessibility in the form of language, availability, and access. Sustainability in all of its forms, is the application or service sustainable, does it hurt or support our environment, the developers, maintainers and users, does it use more or less energy,. Does it support resilience and human well-being? Leave one ingredient out entirely and what you have is at the minimum diminished, and possibly inedible.

Marshall McLuhan argued that every technology is an extension of the human nervous system, a way of reaching further, sensing more, acting beyond the limits of the body alone ([McLuhan, 1964](#)). Universal Cake takes that seriously. If technology extends what humans can do and experience, then the question of who gets to be extended, and who gets cut off, is not a technical question. It is a human one. The signals that extended nervous system carries or does not carry, has a direct effect on the quality of life on our planet. It does not just affect humans, as humans and our technology are in a direct relationship with every being with which we interact.

3. The core idea

Universal Cake is not a checklist. It is an attempt to change what technology is for.

That is worth stating plainly, because it changes how everything else in this document should be read. Most technology evaluation asks: does it work? Does it work fast? Does it work cheaply? Did we hit our target market? These are reasonable questions. But they share an assumption, that the goal of technology is performance, measured against the immediate needs of whoever is already being served.

Universal Cake proposes a different goal: technology that builds, replenishes, and bridges rather than extracts, consumes, and diminishes. Technology that widens participation in its creation and maintenance rather than concentrating it. Technology that supports human flourishing rather than consuming it.

This is not a feature request. It is a fundamental shift, a change to the goal function, to the question the whole system is organised to answer.

When the goal changes, the decisions that follow from it change. The seven dimensions described in section 4 are not a compliance checklist. They are a set of design commitments that follow from a different answer to the question of what technology is for. That is where the leverage is.

Technology is not neutral. Every tool reflects the choices made by its owners, designers, and developers. Those choices affect who can use it, how long it lasts, who controls it, and the effects it has on our world as a whole.

Wellbeing is not a feature. It is a state that emerges from the quality of our relationships, with our bodies, with our minds, and with the people, technology, and world around us. It is reflective of all our relationships with our environment and is also part of that environment. These relationships are not separate. They reinforce one another, or they undermine one another. That is just how it works.

Technology that respects one while degrading another is not a neutral trade-off. It has consequences that ripple outward in ways that are not immediately visible, unlike a feature list or a benchmark score.

Universal Cake proposes a simple shift: stop asking only "does it work?" and start asking "does it work in a way that supports the humans who depend on it? Does it increase or decrease their ability to understand, to participate, to stay well, and to retain agency over their own lives and communities?" That question does not involve a single relationship. It has multiple layers, and attending to all of them, consistently, over time, is what Universal Cake is about.

This shift changes everything. A note-taking app is no longer just fast or slow. It is also asking: can my investment in note-taking survive if the company behind it shuts down, changes owners, or changes the rules of access? Can I share my notes with my whole community if I choose to? Can my blind friends use a screen reader to read them? Will my notes be readable in ten years? Are the notifications designed to respect my attention or exploit it? These are not technical questions. They are human ones, and they always matter more than the technical ones. Applications and data designed to endure are a sound investment, and the expected return on the hard work of building a personal knowledge base and the relationships that surround it.⁴ What Universal Cake evaluates

Universal Cake can be applied to almost any technology, software, hardware, infrastructure, documentation systems, or communication platforms. It evaluates them across a set of human-centered dimensions and needs rather than purely technical ones. No technology will ever score perfectly across all of them. The goal is not perfection. The goal is honest, balanced evaluation, and a growing commitment to doing better by serving better.

These dimensions are not a flat list of equally weighted concerns. They are a map of dependencies. Accessibility is upstream of everything else: without it, the other dimensions cannot be reached at all. Interoperability enables resilience, a system that cannot export its data cannot survive the failure of its host. Transparency enables sustainability, you cannot maintain what you cannot inspect. Knowledge preservation depends on both: knowledge that cannot be exported or inspected will not outlast the system that holds it. Human well-being is not one concern among seven. It is the purpose that gives the other six their meaning.

Accessibility asks whether people can actually use the system, including people who communicate in a different language, people living with disabilities, people using older hardware, people with slow internet connections, and people who communicate in non-verbal languages. Accessibility also asks whether individuals and their communities can afford the application or service over time. This is not a feature to add later. This is the door to the house, and the introduction to everything inside it. Without it, nothing else on the list matters.

Sustainability asks whether the system can last. This includes environmental costs like energy use, but also human costs: does it burn out the people who maintain it? Does it require constant expensive upgrades? Does it serve the individuals and organisations that depend on it, or does it slowly reduce their agency and consume their resources, offering only short-term returns that can be exploited through vendor lock-in, or disrupted by an external actor who, for economic, political, or social reasons, decides to change the rules of engagement?

Interoperability asks whether your data can move freely and whether you can continue using your application or service on different devices. Can you take your work elsewhere if you need to? Does the system use open standards that other tools can read, or does it lock you in, quietly, incrementally, until you have invested so many resources that switching becomes too costly to consider?

Transparency asks whether you can understand what the system is doing and what the agreement with the product or service provider actually is. Can you inspect it? Can you audit it? Or is it a black box that you simply have to trust, and that trusts you with nothing in return?

Resilience asks whether the system, and the data within it, can survive disruption, a company closing, an internet outage, a change in the legal or political environment. Are you still able to access the resources you have created when things go wrong? Can the people who depend on the product or service keep going without an additional, significant investment?

Human well-being asks how the product or service affects the people who use it and the world those people are a part of. Does it reduce or increase their stress? Does it support their healthy habits or undermine them? Does it respect the limited resource of human attention, or does it extract from it? And beyond the individual: does it strengthen or weaken the communities, relationships, and living systems on which human flourishing depends?

A product that improves one person's experience while degrading the conditions everyone shares is not a well-being gain. It is a transfer, from the commons to the individual, from the future to the present, from the many to the few. **Knowledge preservation** asks whether the information stored in the system will remain readable in the future. Will your files still open and be readable in twenty years? Does the system depend on a proprietary format that could disappear, be transformed, or take everything stored in it along with it?

4.1 The dimensions as stocks and flows

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In systems thinking, a stock is something that accumulates over time, knowledge, trust, participation, resilience, immune capacity. A flow is a practice or decision that builds or depletes a stock. The distinction matters enormously for design, because you can change a flow quickly but you cannot change a stock quickly. Stocks have inertia. That inertia is both the risk and the asset.

The most important implication for technology design (and perhaps all design for that matter) is this:

because wellbeing and resilience are stocks, the most important design decisions are not the ones that produce immediate results. They are the ones that consistently build the right stocks over time.

They are the ones that consistently build the right stocks over time. Accessibility is not valuable because it solves a problem today. It is valuable because it builds the stock of participation, which compounds across years and decades. Open formats do not matter because they are convenient now. They matter because they build the stock of preserved knowledge, which will be available to future maintainers whether or not the original creators are still around.

This reframes every dimension of Universal Cake. It is not a compliance requirement. It is a long-term investment logic, a framework for making design decisions that build the right stocks rather than extracting from them.

A technology ecosystem that draws down its stocks of participation, diversity, and preserved knowledge is not efficient. It is consuming its own future.

4.2 Feedback loops within the framework

Three reinforcing loops run through the Universal Cake dimensions. They are worth naming, because a loop you can name is a loop you can consciously accelerate or interrupt.

The knowledge preservation loop. Systems that use open, durable formats preserve knowledge. Preserved knowledge enables future maintainers to understand and improve the system. Understanding enables better maintenance. Better maintenance extends the life of the system. Longer-lived systems accumulate more knowledge. The loop is self-reinforcing in both directions, open formats produce longevity, and longevity rewards open formats.

The exclusion decay loop. This one runs in the wrong direction.

A system that excludes edge communities loses the innovation and resilience that those communities carry

the creative capacity identified in hyphenate status ([Veblen, 1919](#)), the problem-solving capacity found in ecotones ([Odum, 1953](#)), the bridging capital that organisations starve themselves of when they only strengthen existing relationships ([Putnam, 2000](#)).

The system becomes more fragile.

When disruption comes, the fragile, centre-optimised system fails first. Its failure produces more exclusion, as the people who depended on it lose access. The compounded digital exclusion that [Zheng and Walsham \(2021\)](#) document is this loop operating across an entire ecosystem simultaneously.

The bridging capital loop. The Kevin Kintner story in section 2.3 illustrates this directly. Organisations that reach beyond their known supporters, that invest in the edges rather than just harvesting the centre, build bridging capital, which expands the pool of participants, which increases the diversity of perspectives available, which enables better exploration, which builds more bridging capital ([March, 1991](#)). The environmental canvassing story is this loop in action. Name it and it becomes a tool any organisation can choose to run.

5. Why this matters now

Most technology today is evaluated on a short list of criteria: speed, features, price, and popularity. These are not bad criteria. But they are incomplete. They describe what a tool can do right now. They say almost nothing about how it affects the users who adopt it wholeheartedly, how it affects them as they invest their work in supporting or complying with that system, and how it affects them as individual and highly diverse human beings.

Rarely do we ask who gets excluded and what effects that exclusion has on social, economic, and political life. We do not ask what happens when the company behind the tool changes direction, gets acquired, or closes. We do not ask what the tool does to the attention and mental health of the people who use it every day. We do not ask whether the investment in knowledge stored in it will survive the next decade. We do not ask whether the people who maintain it will still be standing

in five years, and we rarely consider the naturally occurring events that could have historically devastating effects, such as solar flares or Carrington-scale events.

The short list optimises for the centre of the bell curve and calls it good enough. Universal Cake does not reject the short list. But it does attempt to prioritise it and place it within the context of the real world, the world in which humans, our communities, and our natural resources are found, the same world in which technology operates.

Universal Cake begins with human technology and human wellbeing. But it does not end there, and that choice is not philosophical decoration. It is a practical necessity. Every systems model has a boundary, a decision about what is inside the analysis and what is outside. Where you draw that boundary determines what feedback loops you can see. The most important feedback loops in this system cross every boundary we might try to draw. The health of the oceans, the health of human communities, and the health of the technology ecosystems those communities depend on are not analogous. They are continuous. The same stressors, cumulative load, immune suppression, exclusion from what is needed to function, operate through the same mechanisms whether the subject is a fish population, a minority community, or an organisation whose knowledge has been locked into a proprietary format. A model that draws the boundary between human and ecological systems will always produce dangerously optimistic forecasts, because the costs of extraction do not disappear when you exclude them from the model. They accumulate outside the boundary and eventually re-enter as catastrophic surprises.

First among the context Universal Cake asks us to hold is language, human languages and dialects in all of their varieties, because without language nothing else can be communicated, preserved, or passed on. Beyond human language sits the vast and largely unrecognised communication of the myriad creatures with whom we share this planet, systems of meaning that predate human technology by hundreds of millions of years. The short list was written for one kind of person, in one kind of place, under one set of conditions. Universal Cake asks what happens when you take seriously the full scale of life on earth, and what our responsibility is to design with that scale in mind, rather than in spite of it.

6. Who Universal Cake is for

Universal Cake is for anyone who uses technology and wants to make better choices about it. That includes individuals choosing a note-taking app, organisations selecting infrastructure, educators building learning environments, and governments procuring public systems.

It is also for builders. Developers, designers, and architects who want to think more carefully about what they are creating, who it serves, and who it leaves behind.

The framework does not require technical expertise to use. Its most important questions are human ones. And human questions belong to everyone.

7. How to use Universal Cake

The simplest way to use Universal Cake is to ask the guiding questions whenever you are evaluating a tool or building a system. Start with language, can people use this in their own language?, and work outward from there.

- Who can use this? Who cannot?

- Does it work in the languages of the people who need it?
- Will this still work in ten years?
- Can I take my data with me if I leave?
- Can I understand what this system is doing?
- Does this system support or undermine the wellbeing of the people who use and depend on it for access to vital and less vital services?
- What happens if the organisation behind it disappears?
- Does it work for people with different abilities, languages, and devices?
- Who is still being left out?

These questions do not always have easy answers. But asking them changes the conversation. They shift the focus from "what can this do?" to "what does this do to people, and to the world they share?" That shift, from capability to consequence, from the centre of the bell curve to its edges, from the short list to the full picture, is the heart of Universal Cake. It is also, it turns out, just good design.

Resources

Universal Cake

- [Christopher Steel, Personal Mission Statement](#)
- [A Systems Thinking Assessment of Universal Cake -- An Introduction](#)

Systems thinking

- [Thinking in Systems](#)

Inclusive design

- [Inclusive Design: The Bell Curve, the Starburst and the Virtuous Tornado](#)
- [Inclusive Design Research Centre, Jutta Treviranus](#)
- [RAPLIQ](#)

Technology and human extension

- [Understanding Media: The Extensions of Man](#)

Digital exclusion and compounded inequality

- [Inequality of what? An intersectional approach to digital inequality](#)
- [The Price of Exclusion, ILO](#)

Edge communities, resilience, and post-traumatic growth

- [Exploration and exploitation in organizational learning](#)
- [Bowling Alone](#)
- [Fundamentals of Ecology](#)
- [The Intellectual Pre-eminence of Jews in Modern Europe](#)
- [Minority stress model](#)
- [Post-traumatic growth inventory](#)
- [Multicultural experience and creativity](#)

Stress, living systems, and the causation problem

- [Ocean Pollution: Effects on Living Resources and Humans](#)
- [Causation in Environmental Law](#)

Language diversity

- [Ethnologue: Languages of the World](#)

Plain language

- [Federal Plain Language Guidelines](#)

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Changelog

Version	Status	Notes
0.2.9	Draft	

Version	Status	Notes
0.2.8	Draft	Applied recommendations from <i>A Systems Thinking Assessment of Universal Cake -- An Introduction</i> (Meadows voice/Steel commission, v0.1.0): (1) section 3 opens with explicit paradigm-shift statement, "Universal Cake is not a checklist. It is an attempt to change what technology is for.", framing the seven dimensions as a goal-function intervention rather than a compliance list; (2) section 4 opens with a dependency map of the seven dimensions establishing accessibility as upstream, interoperability enabling resilience, transparency enabling sustainability, and knowledge preservation depending on both; (3) new section 4.1 introduces stocks and flows as the design principle underlying the framework, wellbeing and resilience as slow-building stocks, the extraction argument made explicit; (4) new section 4.2 names the three previously unnamed reinforcing loops, knowledge preservation loop, exclusion decay loop, and bridging capital loop, with citations carried from the source document; (5) section 5 now states the system boundary explicitly and expands it, framing the human-ecological continuity argument as a practical systems necessity rather than philosophical decoration; secondary CAP anchors added for Veblen, Odum, Putnam, Zheng & Walsham, and March to support new inline citations in section 4.2; assessment document added to references
0.2.7	Draft	[Version reflected in filename; header not updated from 0.2.6 in source]
0.2.6	Draft	Added closing paragraphs to section 2.3: Sindermann fish stress and cumulative stressors; Harvard Law Review causation problem; reciprocal argument that wellbeing cannot be attributed to a single intervention any more than harm can be attributed to a single pollutant; "living systems do not work that way" framing; "cleaning the water" as the closing statement of the section; added Sindermann and Harvard Law Review references and resources
0.2.5	Draft	Author edits for clarity throughout; version bump applied by author
0.2.4	Draft	Replaced section 2.3 placeholder with full text including Kevin Kintner story, March, Putnam, Odum, Veblen, Meyer, Buckup, Tedeschi and Calhoun, Leung and Chiu; added Brown et al. on inflammation; added all new references
0.2.3	Draft	Author edits for clarity; added Catherine Roy and Vie Autonome Montréal; restructured section 2 with subsections 2.1–2.4; Treviranus citation for Pareto naming point; spelling and grammar corrections throughout
0.2.2	Draft	Merged author edits to v0.2.0 with new sections 2.1 and 2.2; updated Inspirations; corrected section 1 voice and Plone narrative; added Eberhard, Pareto, Statista, and Zheng & Walsham references
0.2.1	Draft	Added sections 2.1 and 2.2 on 80/20 critique and compounded digital exclusion; fixed Meadows timing; added Ethnologue, Pareto, Statista, and Zheng & Walsham references
0.2.0	Draft	Complete rewrite of sections 1–7; added Plone WCAG 1.0 AAA history; added Treviranus virtuous tornado and 80/20 critique; added McLuhan; added Three Supports wellbeing framing; added language as first context; added vital services dependency question; updated all references
0.1.2	Draft	Edited to reflect author's actual thinking regarding Universal Cake

Version	Status	Notes
0.1.1	Draft	Replaced Sources and acknowledgements with Inspirations; reframed origin as 1999 checklist; added Meadows, Treviranus, RAPLIQ; corrected CC badge to BY-SA
0.1.0	Draft	Initial draft